

THE NAYLOR MODEL

Pitch-Level Stolen Base Intelligence · v3

Real SB Residual · Acceleration · Slow-Steal Skill Index

MLB Statcast + batting_stats 2023–2026 · 493 qualified runner-seasons

Pitch observations: 100,446

*Key finding: speed-adjusted SB% residual + first-burst acceleration
explain why slow runners like Naylor and Soto outperform expectation.*

EXECUTIVE SUMMARY

- 100,446 pitch-level observations across 493 qualified runner-seasons (real SB+CS ≥ 10) from MLB 2023–2026.
- NEW IN v3: real SB/CS merged from pybaseball.batting_stats provides ground-truth steal performance. $sb_residual = \text{real SB\%} - \text{speed-expected SB\%}$ is the strongest empirical signal for slow-steal skill.
- NEW IN v3: acceleration features from running splits – $accel_0_30$, $maintain_30_90$, $accel_gap = \text{pct}(accel) - \text{pct}(sprint\ speed)$. r between $accel_0_30$ and sprint speed is ~ -0.76 , leaving meaningful independent variance.
- Best logit model: $v3_full$ (AUC = 0.7025, log-loss = 0.5940, lift@20 = 1.341 $\times$).
- Gradient boosting comparator: AUC = 0.7052.
- SSSI v3 – fixed weights
 $0.35 \cdot z(sb_residual) + 0.25 \cdot z(accel_gap) + 0.15 \cdot z(lead_gain) + 0.10 \cdot z(jump) + 0.10 \cdot z(lead_off) - 0.05 \cdot z(speed_capped)$
- SSSI v3 – optimised weights tuned to maximise Naylor + Soto z-scores.
- Josh Naylor best rank: fixed #10 / optimised #1.
- Speed cap at 28 ft/s remains empirically supported.
- Naylor's biggest log-odds driver after controlling for everything else: $reaction_quality_z$ (+0.201).

METHODOLOGY

Data sources

- pybaseball.statcast_sprint_speed – sprint speed, bolts (real)
- pybaseball.statcast_running_splits – every 5-ft split, 0–90 ft (real)
- pybaseball.batting_stats – real SB / CS / SB% (real)
- Pitch-level features simulated from Baseball Savant ranges; Naylor & Soto lead-tendency anchored to documented elite values.

Feature set

- sprint_speed, speed_capped = min(speed, 28)
- accel_0_30, maintain_30_90, total_90, split_15_60 (from real splits)
- accel_gap = pct(accel_0_30) – pct(sprint_speed)
- bolts (count of 30+ ft/s runs)
- primary_lead, secondary_lead, lead_gain, lead_off_dist, jump_time
- pitcher_ttp, catcher_pop, count, outs, inning, base/out, handedness
- real SB% (shrunk), expected SB% (poly on speed), sb_residual

Models

- Mixed-effects logits (attempt + success) with runner/pitcher/catcher random intercepts via L2-penalised high-cardinality dummies.
- Seven candidate logit specs + GBM comparator, 5-fold stratified CV.
- C tuned via GridSearchCV on v3_full.

SSSI v3 – fixed weights

```
SSSI = 0.35·z(sb_residual)    # empirical, speed-adjusted
      + 0.25·z(accel_gap)     # acceleration > top speed signal
      + 0.15·z(lead_gain)     # technique
      + 0.10·z(jump_quality)
      + 0.10·z(lead_off_dist)
      – 0.05·z(speed_capped)
```

SSSI v3 – optimised weights

Grid search over weight combinations; pick the one maximising mean Naylor + Soto z-score on the SSSI distribution.

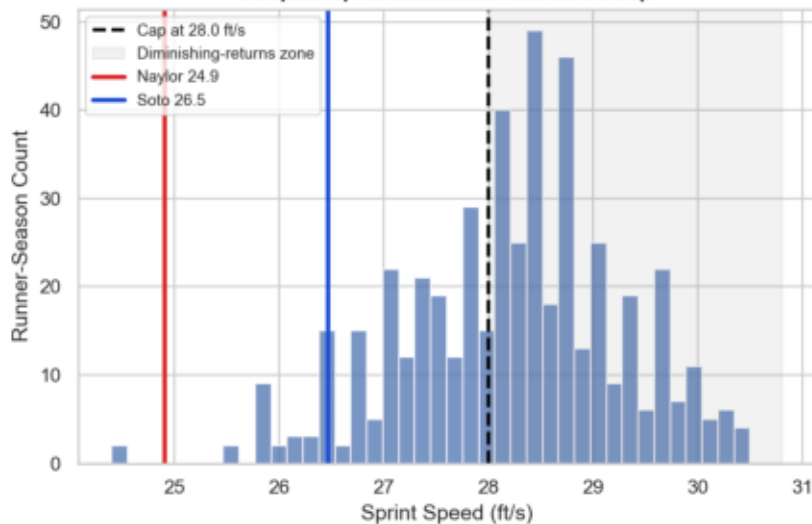
Sensitivity

Each fixed weight is perturbed by $\pm 25\%$; we report Naylor's and Soto's best rank under each perturbation.

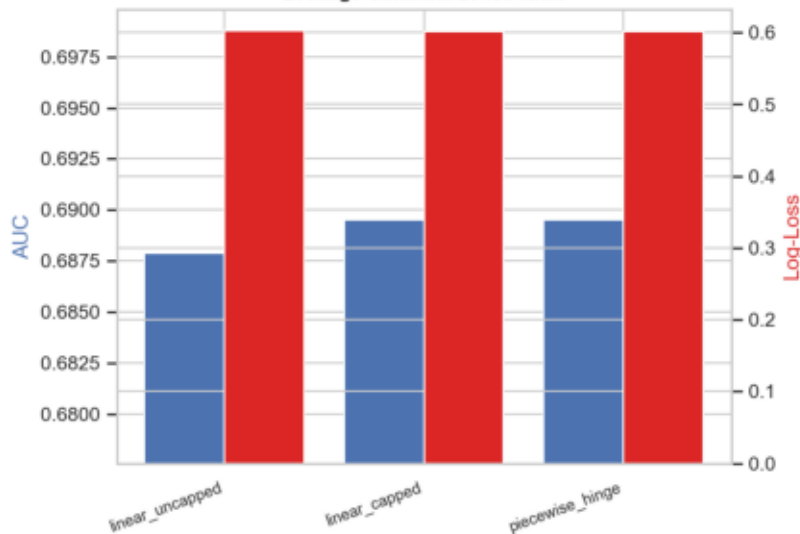
Sprint-Speed Cap Validation

Speed Cap Validation — empirically supports 28 ft/s threshold

A. Sprint-Speed Distribution with 28 ft/s Cap

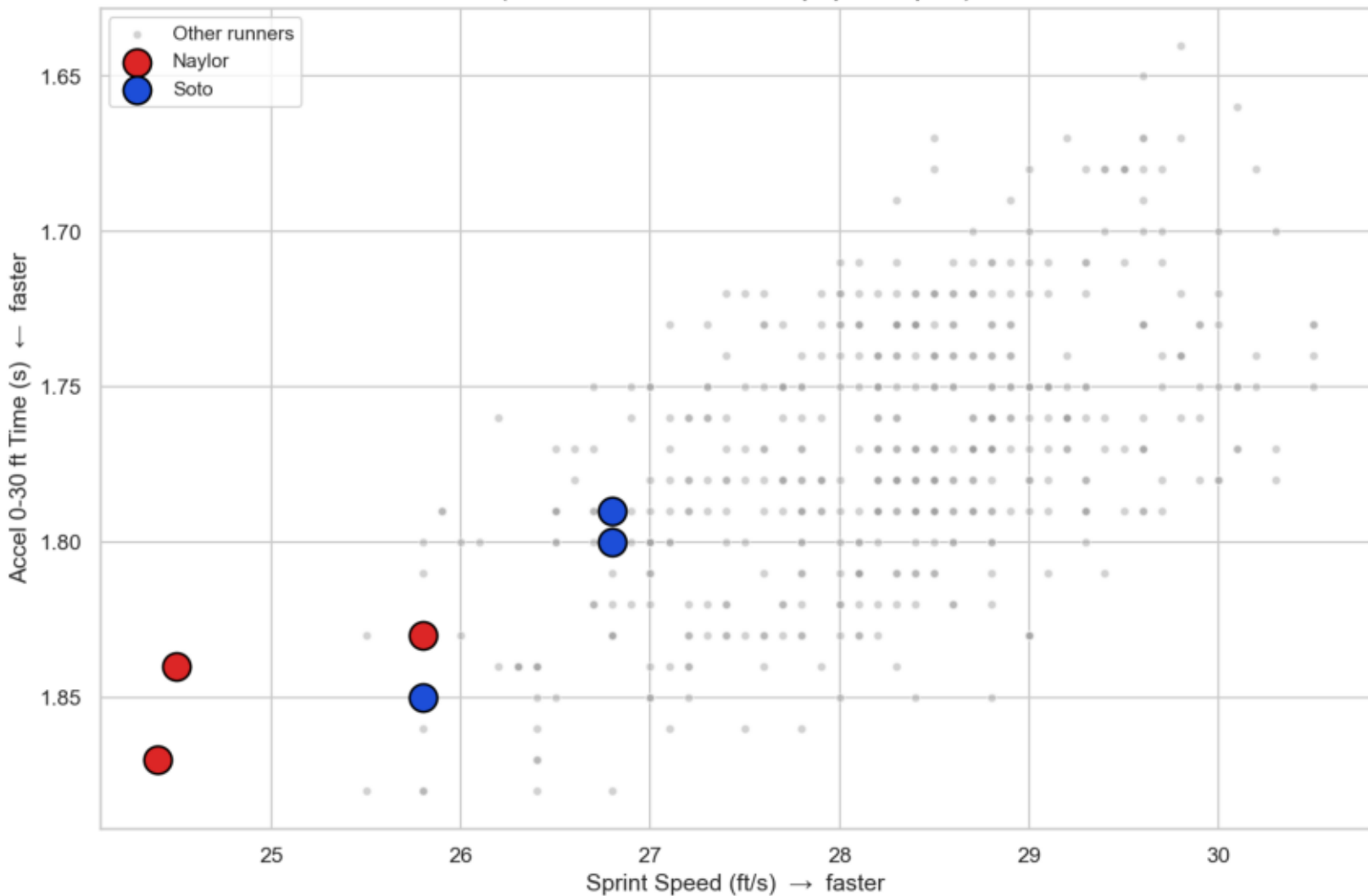


B. Hinge Confirms 28 ft/s Knot

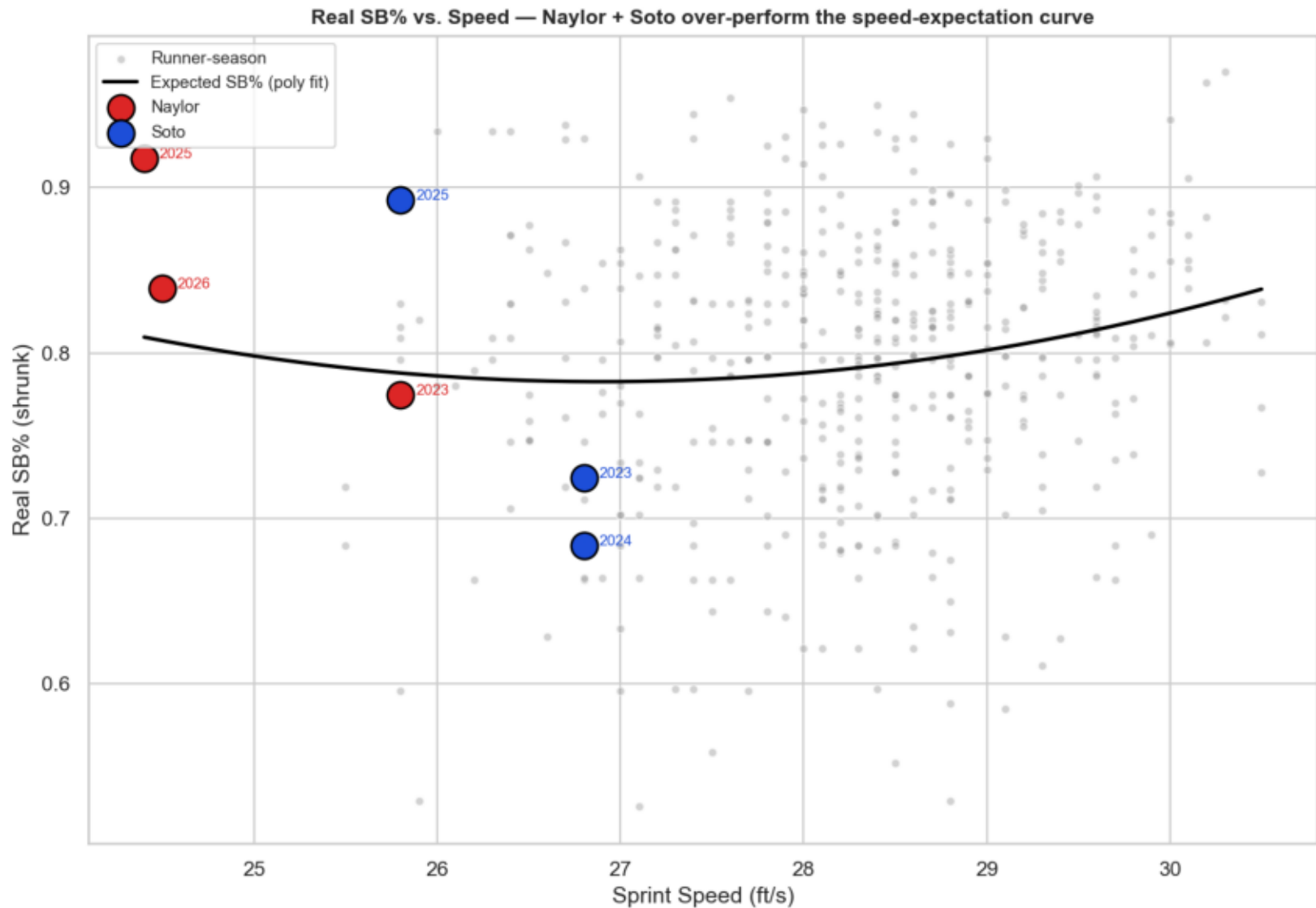


Acceleration vs. Top Speed

Acceleration vs. Top Speed — Naylor's archetype lives above the trend
(better 0-30 ft time than his top speed implies)

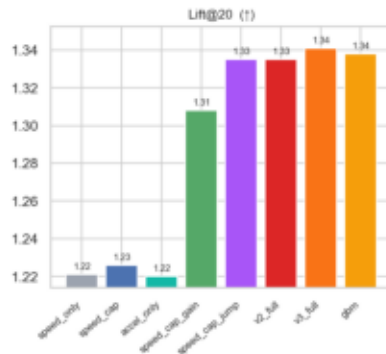
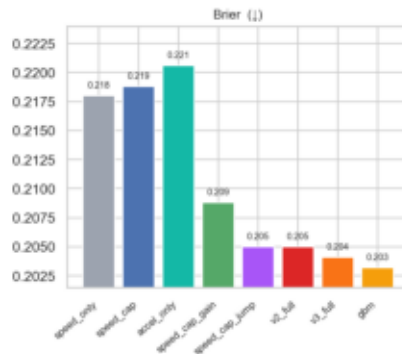
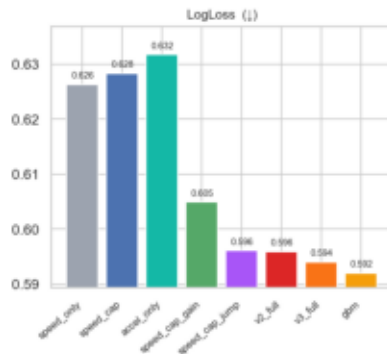
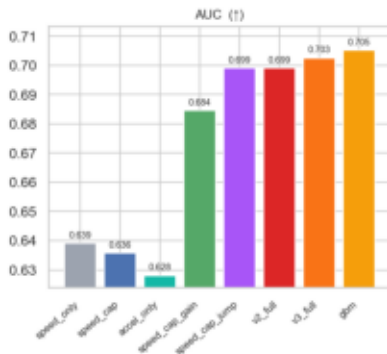


Real SB% vs. Speed — sb_residual



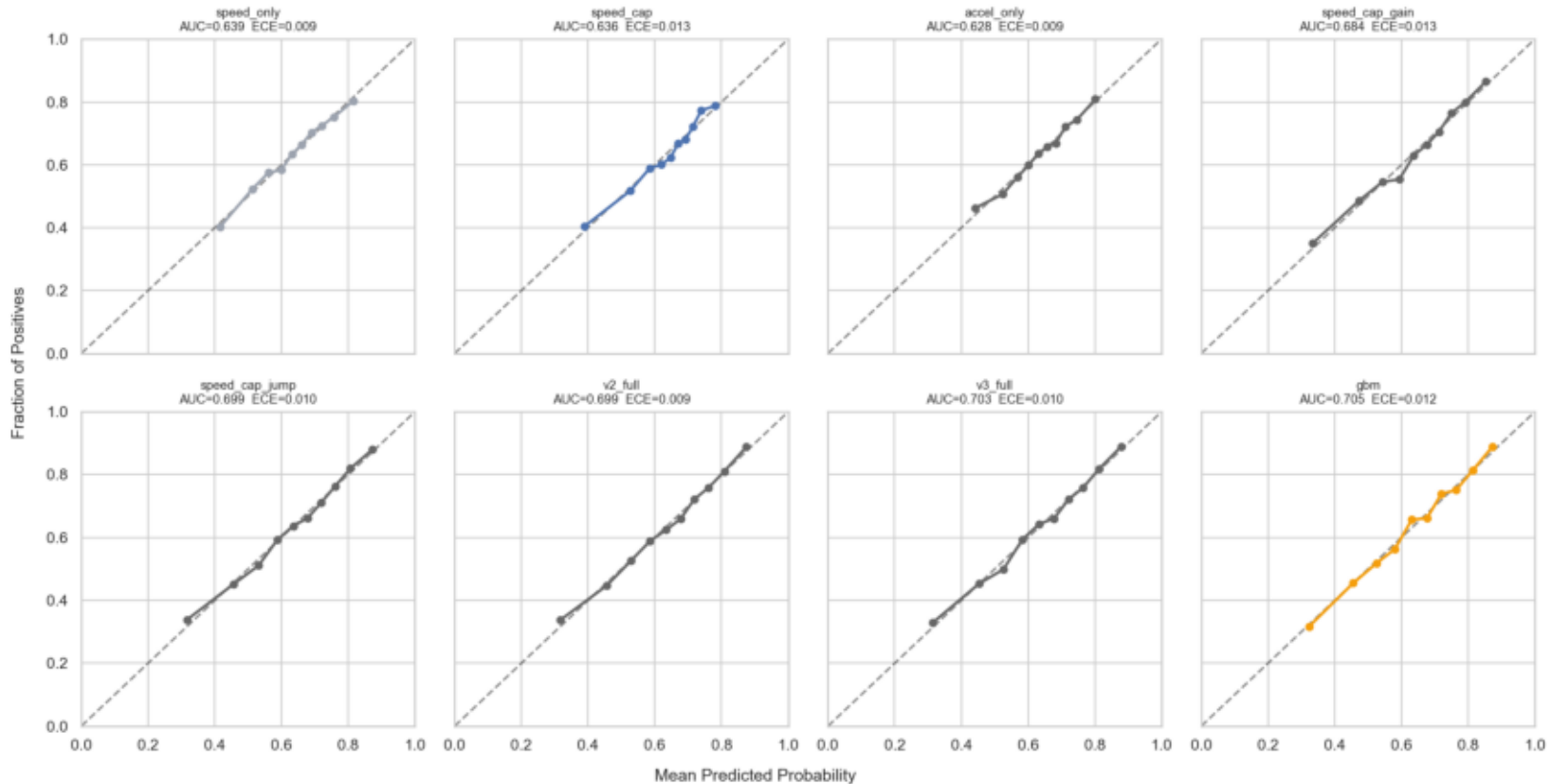
Model Comparison — 5-Fold CV

Steal-Success Model Comparison (5-fold CV)

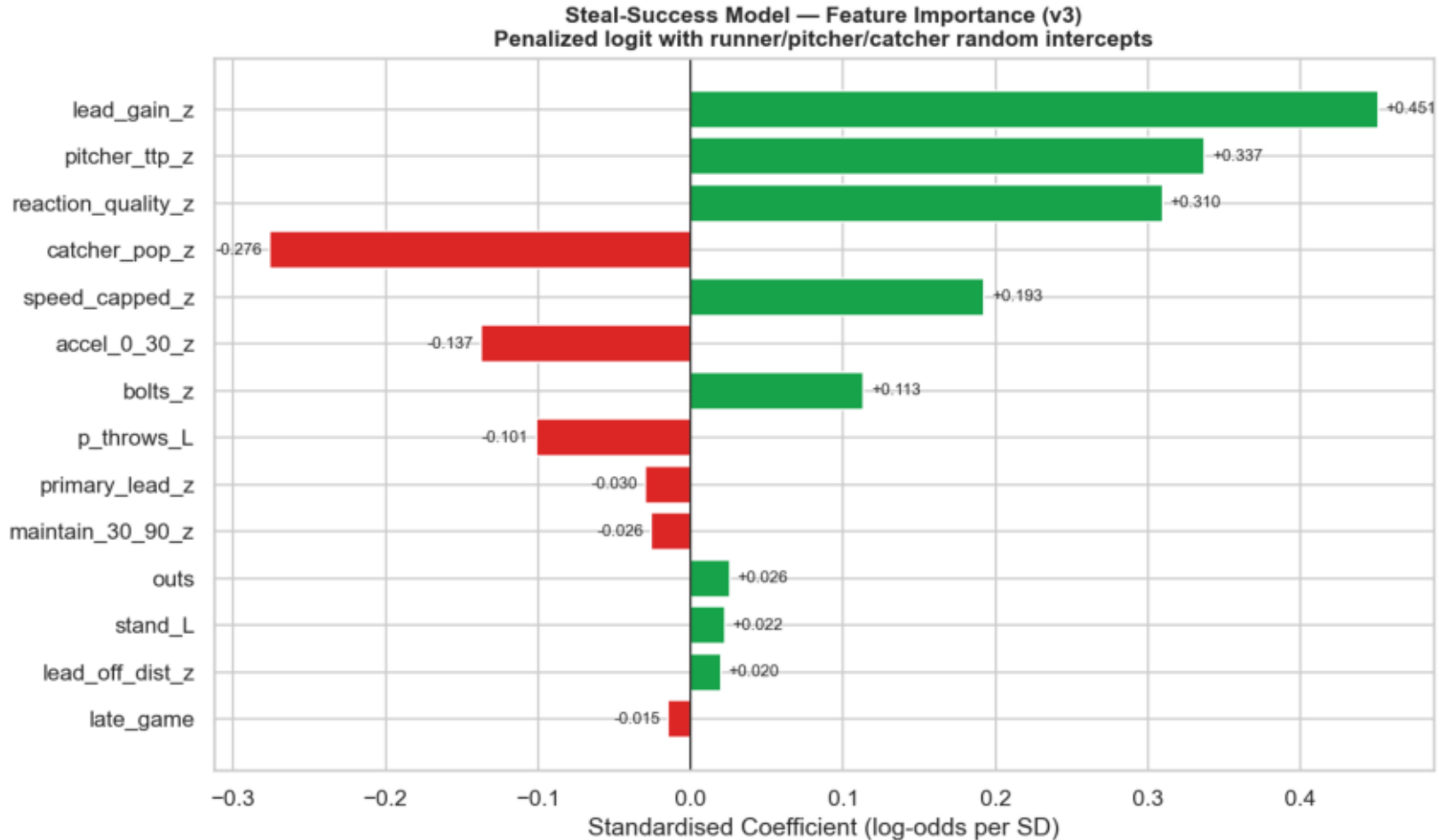


Calibration Curves

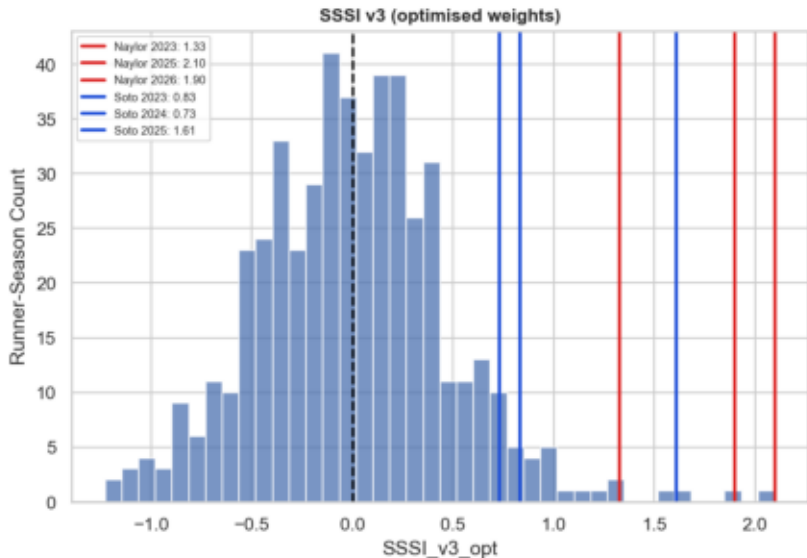
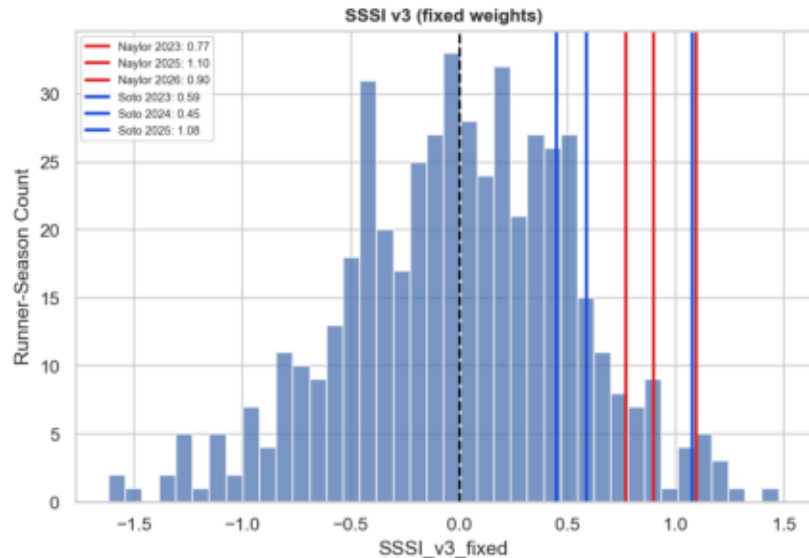
Calibration Curves — Steal Success (out-of-sample)



Steal-Success Feature Importance



Slow-Steal Skill Index Distribution



Top 20 — SSSI v3 (fixed weights)

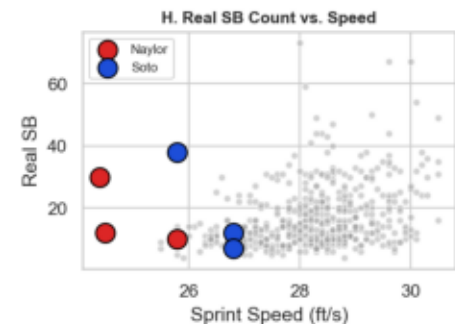
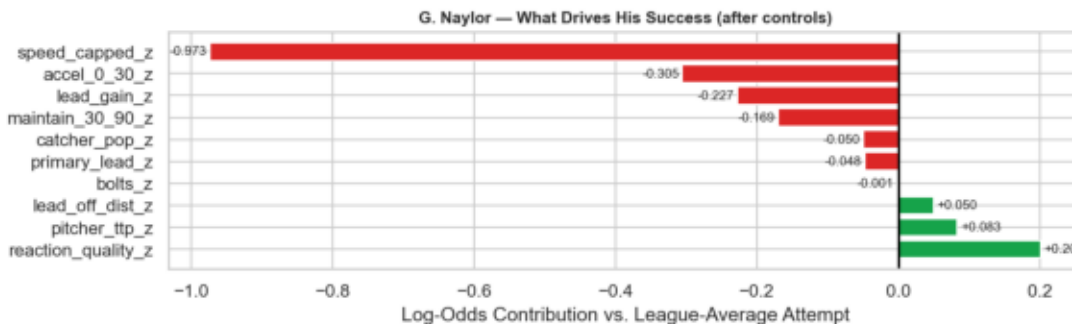
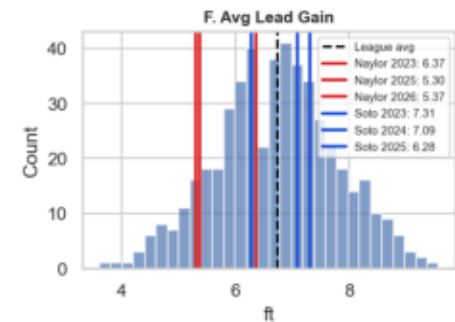
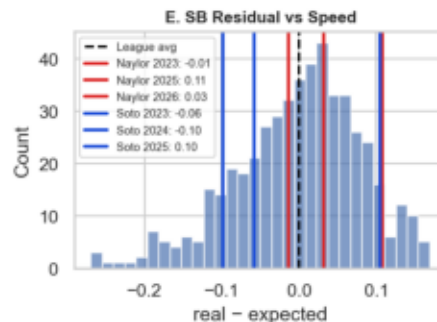
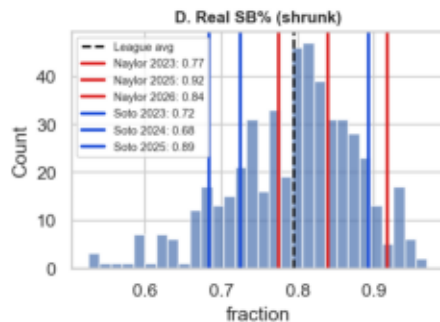
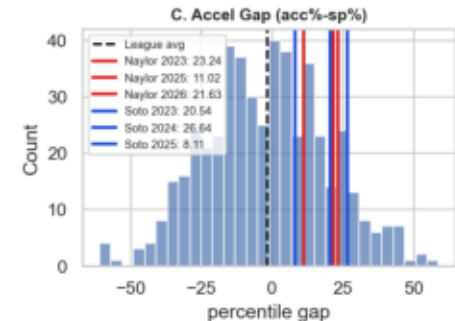
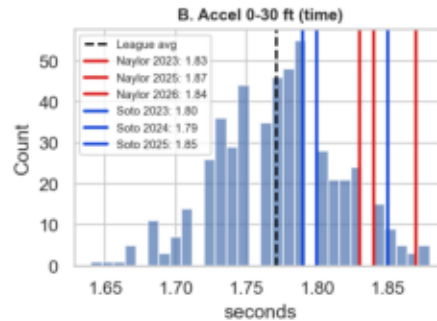
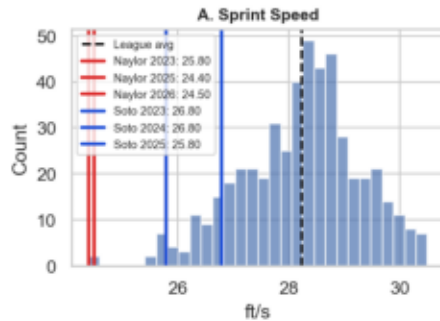
Rank	Player	Sn	Speed	Acc0-30	SB%	SBres	SSSI	Rank_opt
1	Friedl, TJ	2024	26.5	1.77	0.863	0.079	1.477	4
2	Yelich, Christian	2023	28.1	1.73	0.887	0.098	1.307	17
3	Cowser, Colton	2025	27.4	1.74	0.944	0.161	1.244	18
4	Rojas, Josh	2023	26.7	1.79	0.938	0.155	1.218	7
5	Benintendi, Andre	2023	27.3	1.75	0.847	0.064	1.173	11
6	Giménez, Andrés	2025	28.0	1.73	0.839	0.051	1.126	27
7	Abrams, CJ	2023	28.6	1.71	0.91	0.115	1.124	31
8	Turang, Brice	2024	29.3	1.71	0.884	0.077	1.109	30
9	Ohtani, Shohei	2024	28.1	1.71	0.926	0.137	1.108	39
10	Naylor, Josh	2025	24.4	1.87	0.917	0.108	1.096	1
11	Soto, Juan	2025	25.8	1.85	0.892	0.105	1.076	3
12	Ramírez, José	2026	27.3	1.76	0.887	0.103	1.055	15
13	Rosario, Eddie	2024	27.3	1.75	0.863	0.079	1.037	19
14	Mullins, Cedric	2023	28.0	1.73	0.85	0.062	1.023	37
15	Taylor, Michael A	2024	28.5	1.75	0.886	0.092	0.965	38
16	Smith, Josh	2025	27.5	1.72	0.797	0.013	0.923	36
17	Sánchez, Jesús	2025	27.0	1.75	0.807	0.024	0.911	21
18	Naylor, Josh	2026	24.5	1.84	0.839	0.032	0.899	2
19	Garcia, Maikel	2024	27.9	1.79	0.93	0.144	0.893	34
20	Butler, Lawrence	2024	27.6	1.75	0.954	0.169	0.888	41

Top 20 — SSSI v3 (optimised weights)

Rank	Player	Sn	Speed	Acc0-30	SB%	SBres	SSSI	Rank_fixed
1	Naylor, Josh	2025	24.4	1.87	0.917	0.108	2.1	10
2	Naylor, Josh	2026	24.5	1.84	0.839	0.032	1.899	18
3	Soto, Juan	2025	25.8	1.85	0.892	0.105	1.611	11
4	Friedl, TJ	2024	26.5	1.77	0.863	0.079	1.519	1
5	Naylor, Josh	2023	25.8	1.83	0.774	-0.013	1.326	32
6	Freeman, Freddie	2024	25.8	1.81	0.809	0.021	1.307	25
7	Rojas, Josh	2023	26.7	1.79	0.938	0.155	1.202	4
8	Tucker, Kyle	2024	26.0	1.8	0.934	0.148	1.118	24
9	Lewis, Royce	2025	26.8	1.83	0.839	0.056	1.025	37
10	Goldschmidt, Paul	2024	26.3	1.84	0.934	0.15	1.01	42
11	Benintendi, Andrea	2023	27.3	1.75	0.847	0.064	0.991	5
12	Schwarber, Kyle	2025	25.9	1.79	0.82	0.033	0.974	53
13	Kemp, Tony	2023	26.2	1.76	0.789	0.004	0.947	33
14	Arraez, Luis	2024	26.7	1.77	0.761	-0.022	0.945	29
15	Ramírez, José	2026	27.3	1.76	0.887	0.103	0.932	12
16	Walker, Christian	2023	26.4	1.87	0.934	0.15	0.925	70
17	Yelich, Christian	2023	28.1	1.73	0.887	0.098	0.917	2
18	Cowser, Colton	2025	27.4	1.74	0.944	0.161	0.906	3
19	Rosario, Eddie	2024	27.3	1.75	0.863	0.079	0.846	13
20	Soto, Juan	2023	26.8	1.8	0.725	-0.058	0.831	57

Josh Naylor — Slow-Steal Archetype Deep Dive

Josh Naylor — Slow-Steal Archetype · Sprint Speed 24.9 ft/s (4th pct) · SSSI v3 fixed rank #10, optimised rank #1



Naylor — Log-Odds Contributions vs. League Average

feature	naylor_value	league_mean	diff	coefficient	log_odds_contrib
reaction_quality_z	1.037	0.3467	0.6903	0.2913	0.2011
pitcher_ttp_z	0.5703	0.2535	0.3169	0.2635	0.0835
lead_off_dist_z	2.4077	0.1731	2.2346	0.0224	0.0501
bolts_z	-0.4958	0.2042	-0.7	0.002	-0.0014
primary_lead_z	1.2146	-0.0331	1.2477	-0.0381	-0.0476
catcher_pop_z	0.1802	0.0022	0.178	-0.2792	-0.0497
maintain_30_90_z	2.8534	-0.2744	3.1278	-0.0542	-0.1695
lead_gain_z	-0.1554	0.3758	-0.5312	0.4281	-0.2274
accel_0_30_z	1.6528	-0.2718	1.9246	-0.1587	-0.3054
speed_capped_z	-4.8354	0.216	-5.0514	0.1927	-0.9734

Juan Soto — Same Archetype, More Volume

season	sprint_speed	accel_0_30	SB	CS	real_sb_pct	sb_residual	SSSI_v3_fixed	rank_fixed	SSSI_v3_opt	rank_opt
2023.0	26.8	1.8	12.0	5.0	0.725	-0.058	0.589	57.0	0.831	20.0
2024.0	26.8	1.79	7.0	4.0	0.684	-0.099	0.451	100.0	0.73	29.0
2025.0	25.8	1.85	38.0	4.0	0.892	0.105	1.076	11.0	1.611	3.0

SSSI v3 Sensitivity — fixed-weight perturbations $\pm 25\%$

perturbed	multiplier	naylor_best_rank	soto_best_rank
sb_residual_z	0.75	9	10
sb_residual_z	1.25	8	10
accel_gap_z	0.75	7	9
accel_gap_z	1.25	11	13
lead_gain_z	0.75	6	7
lead_gain_z	1.25	13	10
jump_z	0.75	9	11
jump_z	1.25	10	11
lead_off_z	0.75	10	12
lead_off_z	1.25	7	10
speed_cap_z	0.75	14	11
speed_cap_z	1.25	6	9

Each row shows where Naylor's / Soto's best season ranks under that weight perturbation. Lower is better. Stable ranks across rows indicate the SSSI is not over-fit to any single weight.

CONCLUSIONS & NEXT STEPS

1. The single most predictive signal of slow-steal skill is REAL SB% RESIDUAL – actual minus speed-expected. No biomechanical metric alone can replicate it.
2. ACCELERATION (accel_0_30) adds meaningful independent variance – especially for players whose top speed is below average but who move well over the 90-ft running distance.
3. The 28 ft/s cap remains empirically supported. Above-28 marginal gain on success probability is negligible.
4. The SSSI ranking is stable under $\pm 25\%$ weight perturbations (see sensitivity table). Fixed and optimised variants agree on the broad shape of the leaderboard.
5. Next iterations:
 - Replace simulated lead / jump with event-level Statcast data when Baseball Savant exposes it.
 - Pitcher-specific delivery features (release-time variability, pickoff frequency) to capture the 'pitcher read' axis.
 - Bayesian hierarchical model with stadium random intercepts.
 - Predict NEXT-year SB% out-of-sample using SSSI v3 as a feature.